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Unsupervised Machine Learning Methods as a Powerful Tool for Lithological Pattern Recognition Using Wells Logs Data

A. R. Narino, M. Mergany, and A. Alaraifi, Saudi Aramco

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Abstract

The prediction or identification process of lithological facies is one of the pillars of reservoir modeling for any field. However, the lack of representativity of cored data leads to over- or under-prediction of certain types of lithologies, especially because the acquired cored sections hold the best reservoir properties, imprinting bias to the data analysis.

In this study, we will present a carbonate platform reservoir model, constructed by following a sequence stratigraphy framework, using 4th order sequence stratigraphy boundaries for reservoir zonation. Hence, each zone contains both high and low reservoir quality, but with the lack of representativity of low-quality zones, it is necessary to identify a different technique that helps to delineate more accurately the extension of each of them (Fig. 1). This strategy is applied to identify groups of wells that may sustain similar sedimentary features and indirectly detect places where certain changes might occur.

Introduction

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The distribution of the cored wells along the field covers the area as well as uncored wells, but representing only a small portion of the entire reservoir dataset, also affected by the lack of systematic acquisition, containing zones without core data and limiting the accuracy on the definition of environmental or lithological boundaries. As highlighted in Fig. 1, there is a lithological change between wells D and E, but the extension of the such evaporitic body boundary is uncertain to be defined only from the cored wells data.

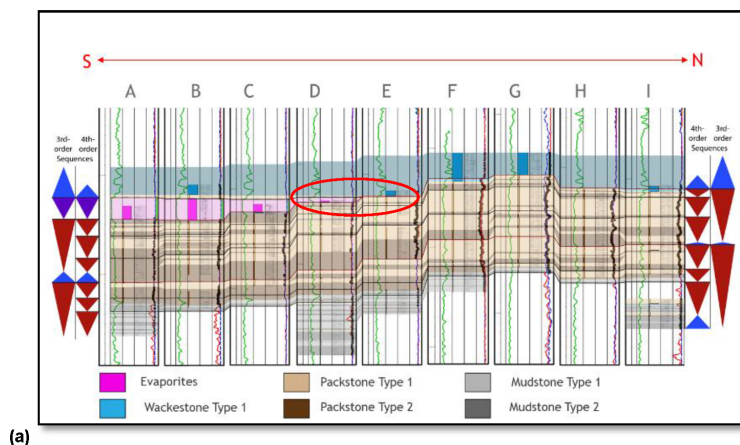


Figure 1—Stratigraphic Concept and Geographic Distribution. (a) Schematic correlation showing the sequence-based stratigraphic architecture, constructed according to the descriptions made from cored wells, applying one of the most common methods of classification of carbonate rocks according to depositional texture (Dunham, 1962). This picture shows how the cored intervals cover mainly the intermediate section of the reservoir, leaving unsampled most of the low-quality intervals at the upper and lower boundaries of the stratigraphic succession.

Objectives/Scope

Often, pre-conceived geological concepts, associated with dispersed and scarce amount of core data, are constrained to high-quality intervals. (Fig. 2) As mentioned before, the lack of systematic sampling and distancing can be compensated by using petrophysical data and extracting additional information from the logs, which leads to the identification of reservoir features beyond facies descriptions. In this work, our thesis is that all the petrophysical logs contain vast amounts of information associated with lithology, which is not fully captured by the core descriptions. The Unsupervised Machine Learning methods are considered as a tool to identify and highlight mathematical relationships without conceptual bias and labeling, allowing to understand the stacking patterns of certain types of lithologies and increasing the confidence in the stratigraphic concept as well as other geological elements that may be controlling the reservoir quality.

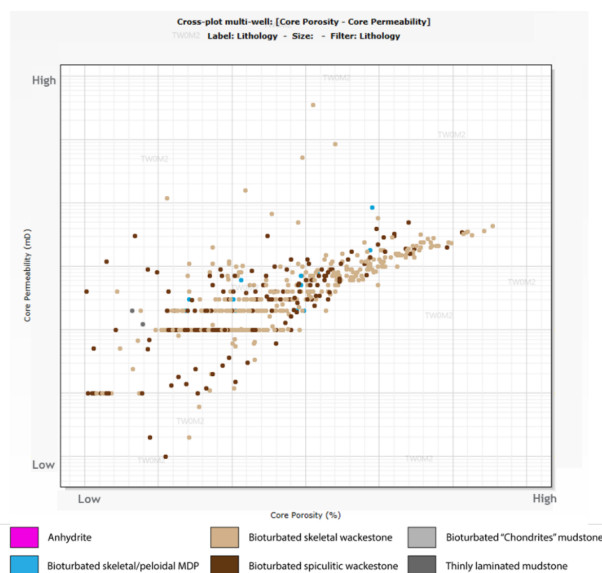


Figure 2—Core Porosity vs. Core Permeability Cross-Plot. In carbonate reservoirs, it is common to find how each lithofacies can exhibit a wide range of properties. Notice how the proportion of the values associated with the packstones type 1 code dominate the population of points in the cloud and observe how the range of permeability covers up to 4 orders of magnitude for a single value of porosity. Also, notice how the sampling of evaporites and mudstones are completely missing in the dataset, as these intervals were not cored or plugged due to their non-reservoir character, resulting in biased outputs.

Methods, Procedures, Process

The process starts by creating an integrated dataset as a source of information. In this study, the core and petrophysical dataset comprises both measured and calculated properties such as neutron porosity, effective porosity, permeability, density, sonic, mineralogical matrix volume contents, gamma-ray and rate of penetration logs, which must be present in all wells. Additionally, conventional core analyses such as core porosity, core density and core permeability are also used for the cored wells. It is determined that each one of these selected log properties or variables must be present in every single well, to maintain dimensional stability and mathematical congruency. For this reason, the PCA (Fig 3.) analysis is crucial to reduce the number of dimensions included in the process to improve the classification accuracy without losing essential features and, at the same time, avoid noise and redundancy. (Padmaja *et al*, 2023)

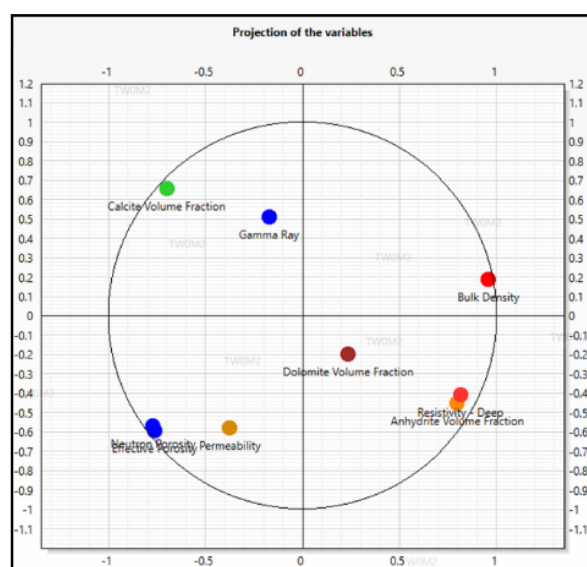


Figure 3—Principal Component Analysis (PCA) is a statistical tool that helps to achieve a dimensionality reduction for the complete dataset composed by elements that are common to each one of the individuals comprised in the study, facilitating the patterns identification among the variables values and to highlighting their relevance.

Regarding the fact that the acquisition of all this information covers a large time span and a wide range of technologies and companies, this dataset needs to be harmonized in matter of naming, definition of units and normalized (Fig. 4) to facilitate the identification of trends among the values. Once it is done, a hierarchical clustering algorithm known as Self-Organized Map (SOM), using the Ward's Method for linkage (minimum variance method), is applied to classify the data into different categories or clusters, computing the distances between points from different groups in a multi-dimensional space. (Hussein, *et al.*, 2020)

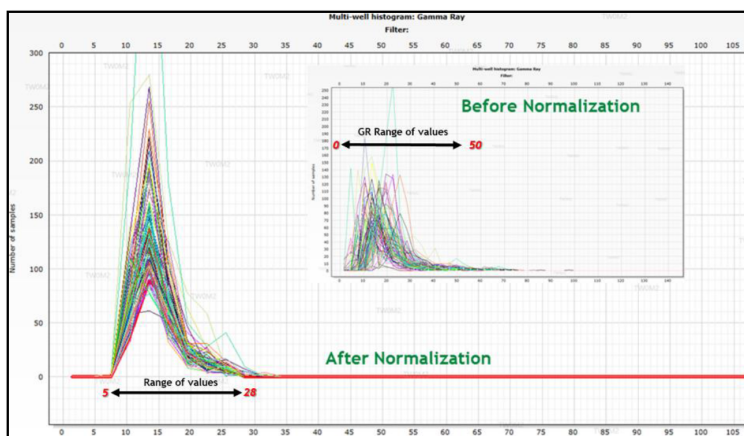


Figure 4—GR Normalization. This is an example of how the normalization of logs is applied to reduce the range of dispersion in data values, offering statistical tools to establish correlations numerically and providing mathematical stability to detect anomalies by reducing the calculated statistical indicators, such as standard deviation, variances and skewness.

One of the objectives of using the unsupervised machine learning method is to allow the algorithm to estimate the number of classes it is capable to extract from the entire dataset in an unbiased manner. To do so, tests are done iteratively, increasing systematically the number of classes expected for each test and studying the resultant correlation coefficients according to the PCA elements contained. (Fig 5.)

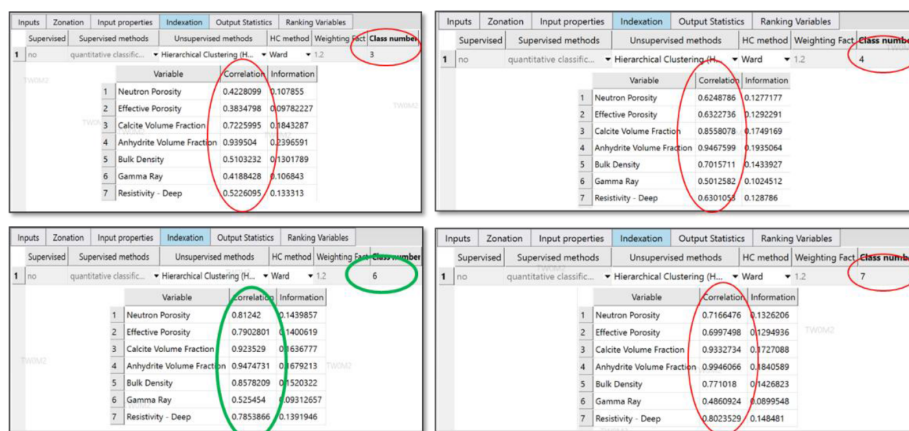


Figure 5—Unsupervised Learning iterations. Different attempts need to be done systematically, maintaining the main hyperparameters unchanged as a background, while modifying only the relevant output. In this classification study, the sole purpose is to understand how many classes can be identified with the maximum grade of correlation. As a result, it is expected to achieve a better segmentation within the dataset.

When the highest correlation coefficients are achieved, the selected clustering is applied to wells, which are further organized geographically to detect spatial trends and vertical distributions. Finally, core descriptions are integrated into the classification to identify whether there are effective textural differences between the samples and then define geographical boundaries for each group of wells. The goal is to understand the relationship with petrophysical and/or dynamic data, linking the unsupervised classification with a reservoir architecture that shows consistency with the given geological concept.

Results, Observations, Conclusions

Core descriptions combine textural and compositional elements, mainly linked to depositional environments. However, this product is not enough to reproduce the dynamic data, flow conditions, or saturation behavior. On the other hand, the unsupervised learning method is becoming popular for

lithotype classification, as they eliminate labeling bias and facilitate the correlations between logs and core information, revealing hidden patterns that can be used to improve reservoir characterization.

Often, sequence stratigraphic modeling is independent from the traditional petrophysical approach of rock typing, which is mainly focused on the identification of reservoir flow properties, regardless of the sedimentary architecture and its genetic relationship. It is important to mention that, even though the petrophysical rock typing exercise is the best tool to parametrize flow simulations, we believe that providing a more accurate delineation of the stacking patterns and their boundaries will help to replicate the flow conditions with a higher precision, introducing stratigraphic elements that might clarify unexpected flow anomalies observed in the dynamic data, with the objective of achieving a more realistic representation of the reservoir fabric within the property models. Additionally, this approach can reveal additional correlations that highlight missing components within the geological concept.

This methodology has been widely used in geophysical data segmentation to use the seismic amplitudes in the delineation of geological boundaries and the prediction of unknown subsurface structures (Couleau, *et al.*, 2003 and Roy & Marfurt, 2011). However, in this study, the main goal is the grouping of wells that share similar stratigraphic or environmental conditions. In Fig. 6a, it is shown that after organizing the wells in N-S direction, it is possible to identify how the lithologies present in the lower interval of the reservoir do not extend unchanged from north to south, but they are somehow truncated towards the central region of the field. Also, It is evident how the upper section in the south contains lithological features that are not present anywhere in the central and northern regions. Likewise, by observing the vertical stacking (Fig. 6b) before applying the proposed sequence boundaries to the correlation, it is evident how certain classes are repeated in different intervals, but in different geographic positions, which shows a resemblance to the natural cyclicity in the environmental changes that shape the stratigraphic succession architecture.

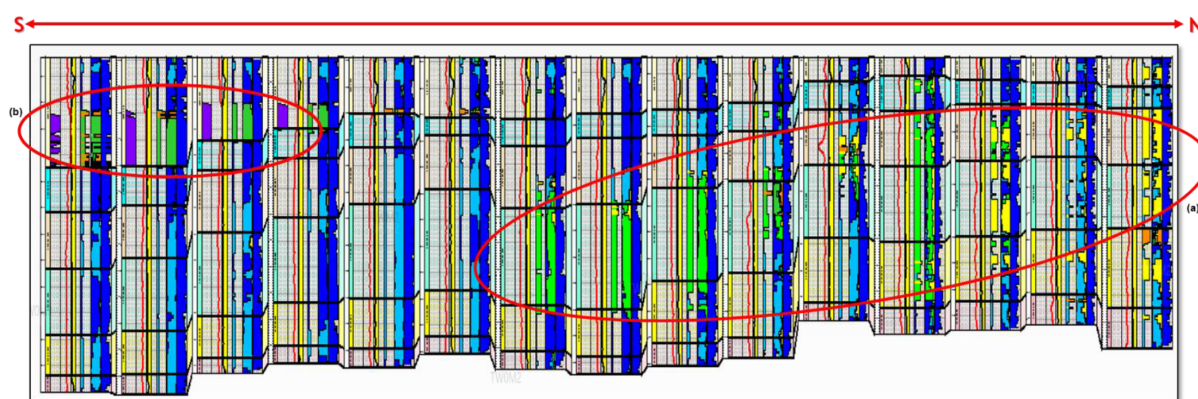


Figure 6—Unsupervised Learning Classification (S-N Correlation). Here it is shown how, once the unsupervised classification is distributed geographically, it is possible to detect spatial trends that highlight how the petrophysical record responds to the reservoir architecture, including both cored and uncored data, hence facilitating the interpretation of reservoir bodies continuity and connectivity as well as their relationship with the stratigraphy, validating the accuracy in the position of the different sequence boundaries. (a) The area within the ellipse highlights the lack of continuity towards the south of rocks clustered under classes 3 (yellow) and 4 (green), which are not present in the middle and lower part of the reservoir, unlike what is observed in the northern region where they are the dominant classes. (b) Notice how class 6 (purple) is exclusively present in the upper part of the reservoir towards the south, which according to the sequence stratigraphic concept, corresponds to the evaporites zone. The extension of this zone is now clearer, thanks to the significantly increased amount of information resulting from the combination of cored and uncured wells in the learning process.

These qualitative observations allow the geographical division of the wells in four genetically associated groups, where the unsupervised learning delivered six classes of elements distributed along the field and across the reservoir. Thanks to this spatial segregation, it is possible to identify not only that all the classes are present in all the areas, but also how different the vertical distribution is from one group to the adjacent one. (Fig 7.)

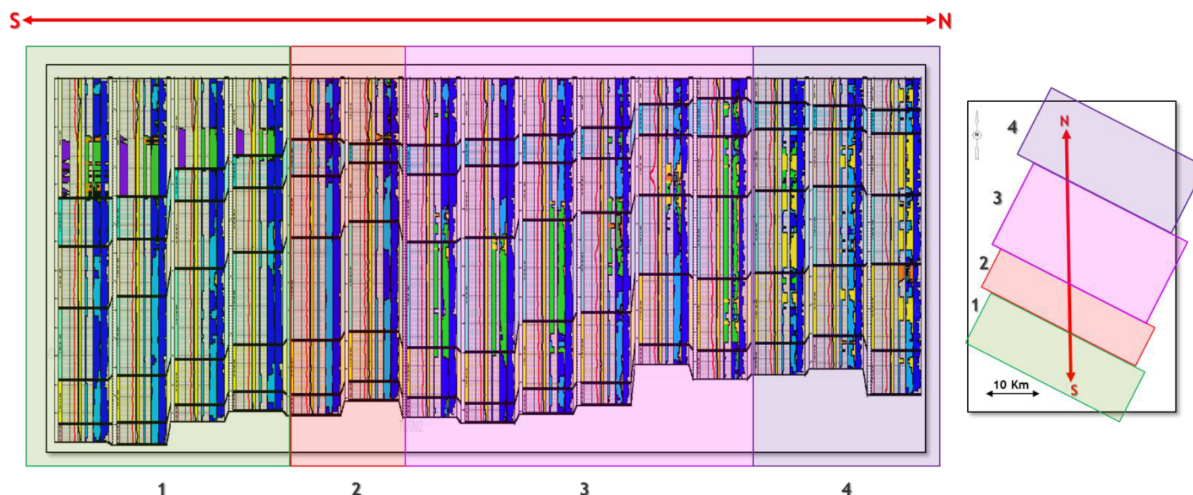


Figure 7—Geographical Groups Distribution. Here it is evident how, by using the unsupervised classification results, it is possible to identify both lateral and vertical changes in the reservoir. This grouping allows to segregate specific classes which are not present in all the groups, placing together wells with similar vertical signatures and predicted classes, and proposing transition or truncation zones that could provide more robustness to the sedimentary model and permit the detection of repetitive features that might be used to validate or modify the position of the initially interpreted sequence boundaries.

To validate the pre-existing interpretation, the next step is to integrate the unsupervised classification with the core descriptions and the interpreted sequence boundaries to demonstrate (or not) its alignment with the stratigraphic conceptual model. In this case, many of the environmental limits are approximately well-captured, but others are not clear, and the presence of unexpected lithological bodies in areas where the sedimentary maps predict a different configuration, permitting the refining of such environmental maps to include these elements more consistently (Fig 8.)

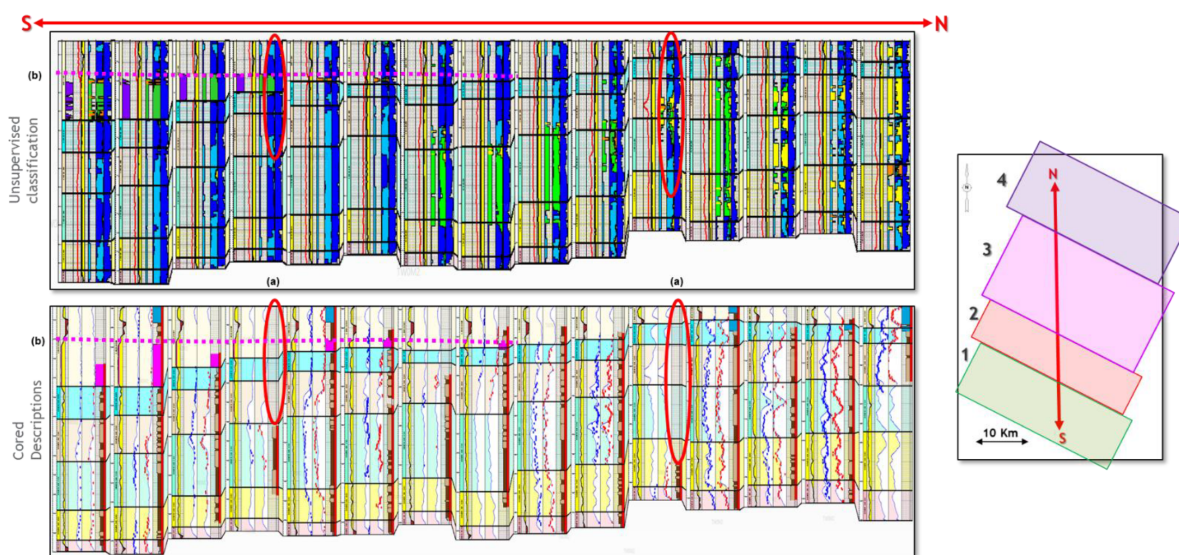


Figure 8—Unsupervised Classification vs. Core Descriptions. In this comparison between the unsupervised learning and the core facies classifications there are several differences, but we can focus mainly on two of them. (a) The first feature to highlight is the complete coverage along each well for all the zones, whereas there are missing sections in the cored interval, reducing the training capability for the supervised (labeled) learning in certain zones, sometimes within the same group. (b) Having the wells organized geographically eases the correlation and identification of missing parts in the interpretation. The dashed line shows how the top of the evaporites may induce substantial differences in the geological concept, depending on how it represents the position of a missing sequence boundary that can describe either a lateral heterogeneity or an undetected unconformity, which will require additional study and validation. Also, it could represent just a paleo-topography effect that can impact vertical connectivity within the reservoir with no impact on the position of the adjacent sequence boundaries as timelines.

This, as any other reservoir characterization exercise, aims to create a 3D reservoir model. However, it is well known that the carbonate-dominated reservoirs exhibit wide ranges of properties for each single category of facies, making it very difficult to correlate textural descriptions (core data only) with petrophysical rock types (logs data only) to validate the facies-based models. However, if we can combine both groups of information into a single dataset, the resultant clustering can work for geologists, modelers and engineers at the same time, to simulate the behavior of the fluids across the reservoirs with full representativity of the dynamic data and the geological concept. In Fig. 9, we can observe how the qualitative separation of each class is easier to define and finally, when comparing the resultant unsupervised classes with the provided facies codes, additional facies associations, or just facies lumping, which can be applied to develop new classes, segregating the lithological bodies with higher accuracy, according to their dynamic behavior, but respecting the sedimentary sequential configuration and its intrinsic architecture. In summary, rock-typing and facies predictions are enriched. A more realistic reservoir architecture, adjusted to the needs of the model, fluid characteristics, and appropriate level of complexity, simplifies the understanding of flow conditions and the identification of additional elements affecting reservoir productivity but maintaining geological coherence.

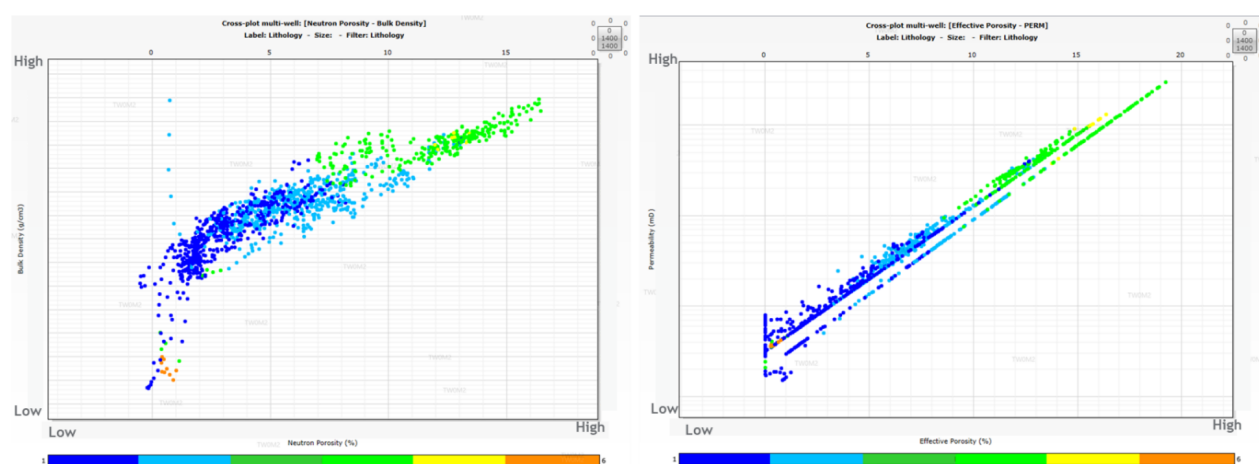


Figure 9—Petrophysical Data Cross Plots. By using the traditional cross-plotting techniques to establish numerical correlations and trends, it is possible to differentiate the major reservoir qualities. Now, by applying the unsupervised classification results, we can observe the separation between the assigned classes beyond the 2 or 3 classes that can be deduced from the traditional approach. Here, it is visible that there are more complex interactions among the classes, rather than just applying thresholds or fixed cut-offs, as in a rock-typing exercise. In this example, there is a very narrow window for class 6 (orange dots) in the Neutron - Density cross plot, whereas according to the calculated effective porosity we can find that there is a complete overlap between low quality class 1 (blue dots) and class 6 (orange dots). This means that regarding its quality, the geomodeler can implement facies lumping at the south region that captures both, the presence of the evaporitic layers and the low porosity and permeability found in the lower mud-dominated section without affecting the facies model or the stratigraphic framework.

Novel/Additive Information

This study reveals that the numerical correlations between the elements in the dataset, represented in the physical response of the formation in the petrophysical logs (log signatures), depends completely of its internal structure, lithological character, or chemical composition, and such features may or may not be present in the cored intervals. By combining these independent elements and leveraging the power of the unsupervised learning machine algorithms to detect elements, correlations, trends and other mathematical interactions, we observed that the descriptive and interpretative tools provided by the traditional geological approaches can be refined, modified or challenged to go beyond the limited amount of data acquired by the logs or the cores separately.

It is common in petrophysics to ignore how the spatial arrangement of the individual datasets effectively impacts the interpretation of the rock typing results. It means that a certain rock type can be present anywhere

with no other physical impact. However, from the predictive point of view, even though the hydraulic behavior is similar, the geological meaning is completely different, impacting the extent of the porous body, the efficiency of the flooding, the quality of the seals at the top of the reservoir, and many other elements that can only be understood if the reservoir architecture is accurately represented.

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